

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

AI Engineer specializing in retrieval-augmented generation, multi-agent orchestration, and workflow automation using Python, GCP and Azure. Experienced building end-to-end ML pipelines, deploying agentic systems with LangChain and HuggingFace, and designing model evaluation frameworks. Researched domain-adaptive ensemble learning for medical nucleus detection (Faster R-CNN, Mask R-CNN, YOLO) at Stevens Institute.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, PySpark, TypeScript
- **AI & Machine Learning:** Prompt Engineering, Model Evaluation & LLM Evaluation (RAGAS), Retrieval-Augmented Generation, Multi-Agent Systems, Model Deployment, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, FastAPI, Embedding Models, Scikit-learn, ChromaDB, Modern Large Language Models (LLMs), Modern AI Frameworks and APIs
- **Infrastructure & DevOps:** AWS Deployments, Google Cloud Platform, BigQuery, Azure Services, Docker, PostgreSQL, Relational Databases, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc

Jun 2025 - Aug 2025

Data Science Intern

CA, US

- Built end-to-end machine learning data ingestion pipelines using Python, PySpark, GCP, BigQuery, and scikit-learn, automating the processing of 200+ MP4 files daily, developing analytical frameworks that reduced validation latency by 40% and enabled data-driven operational decisions at scale
- Deployed Streamlit and Superset-based real-time observability stack over 100K+ records with Dockerized MLflow model tracking, HuggingFace model monitoring, ChromaDB integration, and FastAPI endpoints, improving production ML workflow reliability by 60% and enabling proactive incident response for business-critical pipelines
- Collaborated with product partners to integrate LangChain-powered LLMs into data ingestion pipelines, enhancing real-time insights and improving decision support

VIEW Synergy

Jan 2024 - Apr 2024

AI and Automation Intern

KA, IND

- Architected and deployed AI-driven workflow automation pipelines integrating ReactJS, TypeScript, RESTful APIs, Azure cloud services, LangChain, and LlamaIndex, enabling intelligent task orchestration and reducing manual process overhead for 100+ employees across full SDLC delivery cycles
- Developed scalable microservices-based backend systems using C#, .NET 8.0, SQL, Docker, and the OpenAI API, implementing retrieval-augmented generation pipelines to deliver production-grade, low-latency API serving within an Agile SCRUM delivery model, accelerating cross-functional workflows and driving reliable release cycles
- Designed and built end-to-end AI-centric data workflows using Python and PostgreSQL, ensuring optimized performance and reliability

PROJECTS

RAG Enterprise - SEC EDGAR Filing Analyzer

- Built and evaluated a production RAG system over SEC 10-K filings (AAPL, JPM, TSLA, MSFT, GS) using Python, LangChain, LlamaIndex, BeautifulSoup, and ChromaDB; defined metrics for faithfulness, relevancy, and recall, improving scores from 0.57/0.58/0.57 to 0.65/0.68/0.70 through iterative retrieval tuning
- Enforced investment-banking-grade RBAC, MNPI guardrails, and an SEC 17a-4 / FINRA 4511-aligned audit trail at the vector-store layer; deployed via Docker, Terraform-provisioned AWS ECS Fargate, and GitHub Actions CI/CD.
- Implemented an evaluation pipeline using SQL in PostgreSQL to store RAG metrics, integrating GPT-4o-mini for automated assessment and enabling continuous performance monitoring

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Built a production-grade multi-agent orchestration platform using LangGraph, GPT-4o, and Claude Haiku with a 3-layer agent hierarchy (Supervisor, Specialists, Reviewer), parallel task execution, and a two-tier memory system (Redis + ChromaDB) that improves planning accuracy across sessions - deployed via Docker Compose with Celery workers, a FastAPI backend, and a Streamlit observability UI with cost tracking, step-through replay, and full execution tracing.
- Implemented a Human-in-the-Loop escalation system with 5 automated triggers (low confidence, repeated failure, sensitive operations, low review score, user request), a real-time approval queue with chat-based clarification, and 4 granular approval levels-enabling autonomous completion of ~85% of tasks while routing the remaining 15% to human reviewers with full audit trail, satisfying the compliance requirements of regulated industries such as fintech and legal.

WorkFit

- Developed a real-time multimodal posture monitoring system using CNN-based computer vision and flex sensor fusion, achieving 40% discomfort reduction and 85% user adoption, recognized with the Best Research Paper Award at ICICC 2024 for real-world impact and technical novelty.
- Designed and trained a deep learning inference pipeline for simultaneous eye-strain and postural deviation detection, optimizing for low-latency real-time predictions on resource-constrained hardware.

EDUCATION

Stevens Institute of Technology

May 2026

Master of Science, Applied Artificial Intelligence

NJ, US

PES University

May 2024

Bachelor of Technology, Computer Science (Specialization in ML)

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